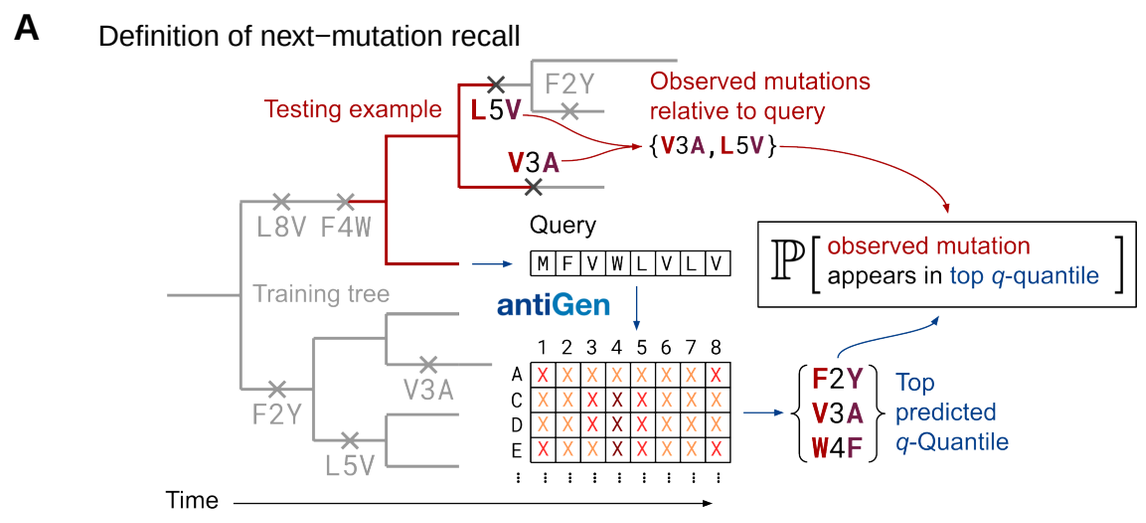
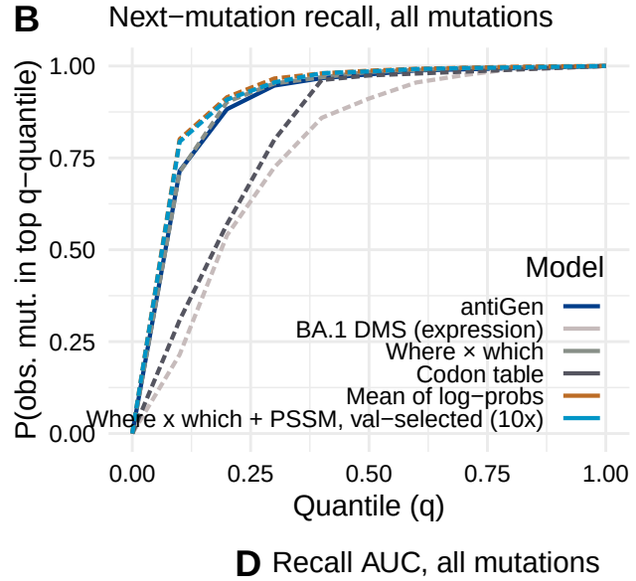
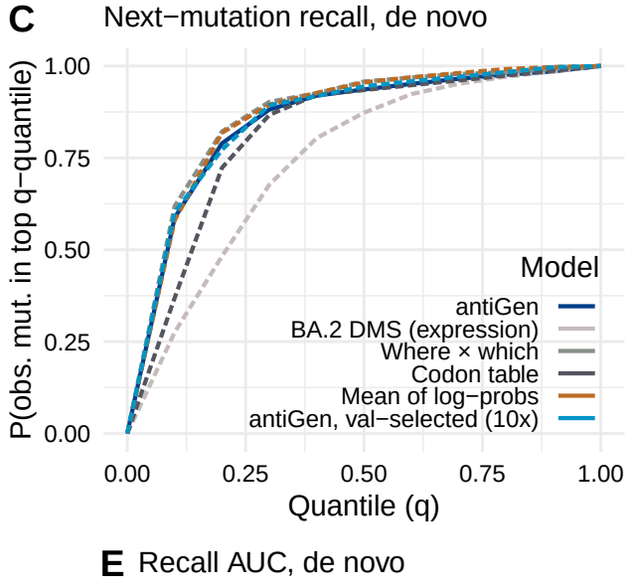
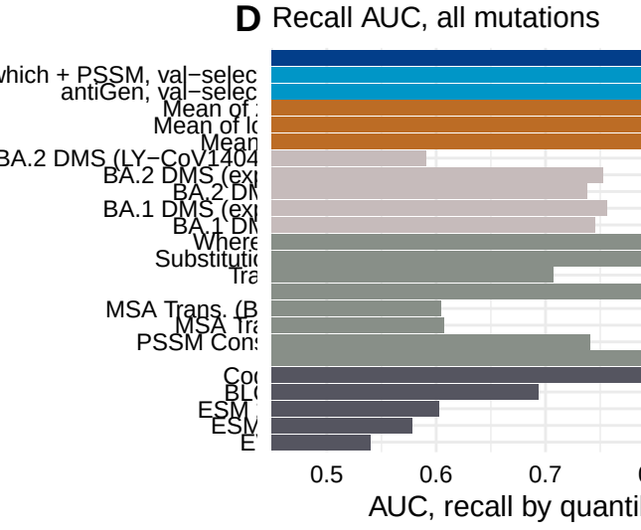
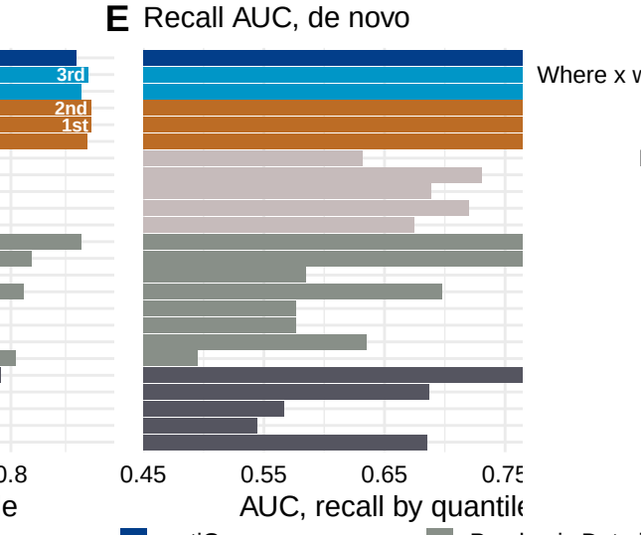

A
Definition of next−mutation recall
Testing example
L5V
F2Y
V3A
Observed mutations relative to query
{V3A, L5V}
L8V F4W
Query
M F V W L V L V
Training tree
antiGen
1 2 3 4 5 6 7 8
A
C
D
E
ℙ[observed mutation appears in top q-quantile]
F2Y
V3A
W4F
Top predicted q-Quantile
Time

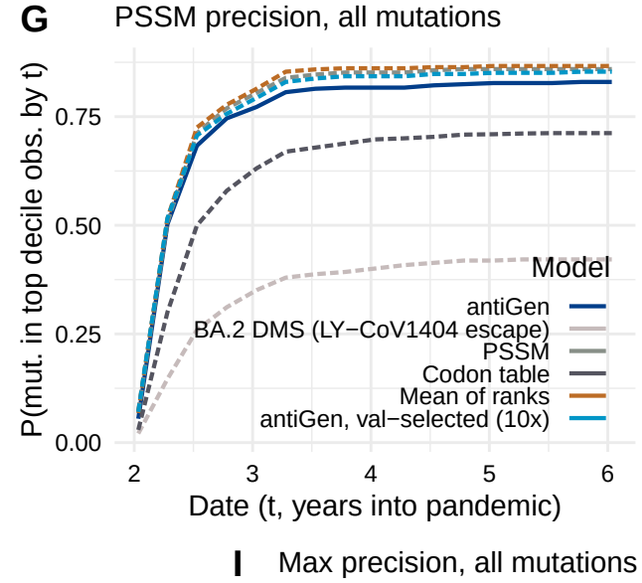

B
Next−mutation recall, all mutations
P(obs. mut. in top q−quantile)
Quantile (q)
Model
antiGen
BA.1 DMS (expression)
Where × which
Codon table
Mean of log−probs
Where x which + PSSM, val−selected (10x)

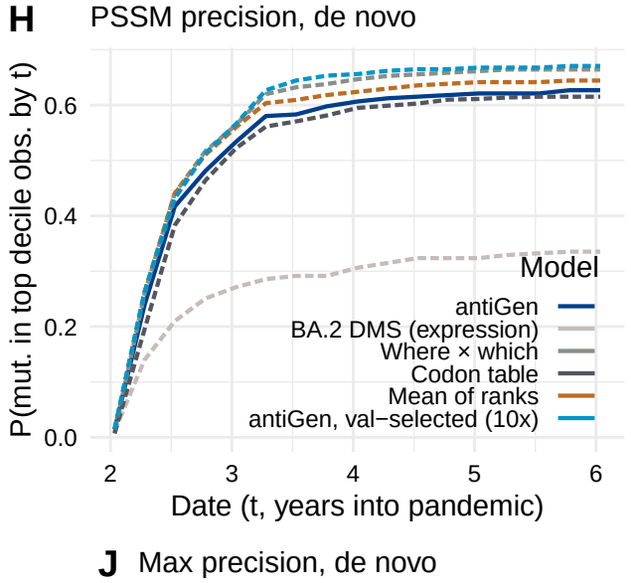

C
Next−mutation recall, de novo
P(obs. mut. in top q−quantile)
Quantile (q)
Model
antiGen
BA.2 DMS (expression)
Where × which
Codon table
Mean of log−probs
antiGen, val−selected (10x)

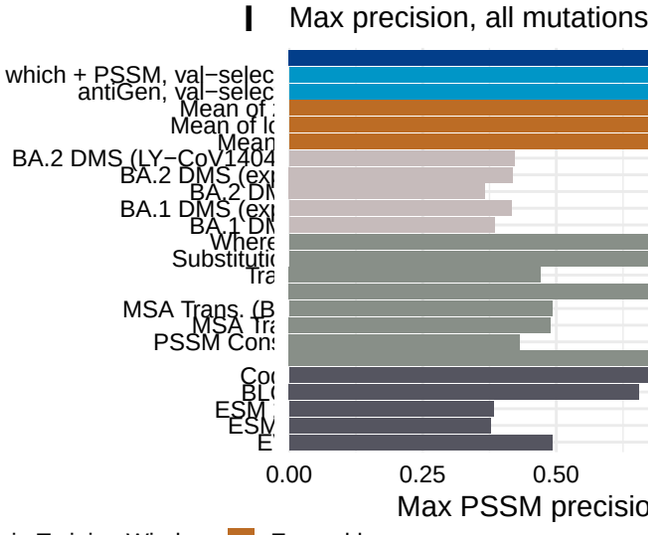

D
Recall AUC, all mutations
Where x which + PSSM, val−selec
antiGen, val−selec
Mean of
Mean of lo
Mean
BA.2 DMS (LY−CoV1404
BA.2 DMS (ex
BA.2 DM
BA.1 DMS (ex
BA.1 DM
Where
Substitutio
Tra
MSA Trans. (B
MSA Tra
PSSM Cons
Co
BL
ESM
ESM
E
3rd
2nd
1st
0.5
0.6
0.7
0.8
AUC, recall by quantile

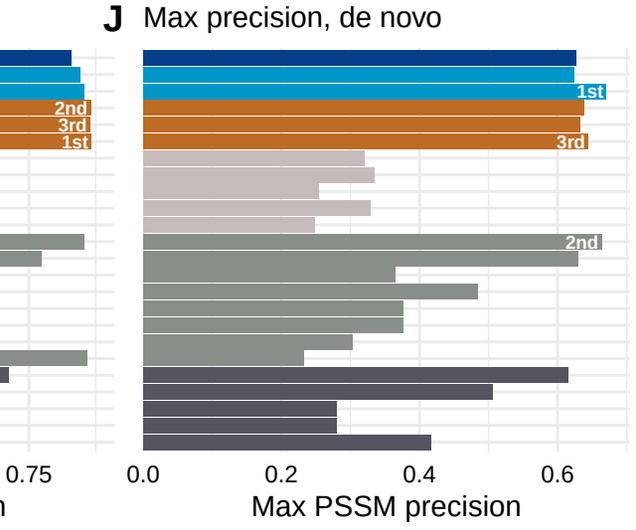

E
Recall AUC, de novo
0.45
0.55
0.65
0.75
AUC, recall by quantile

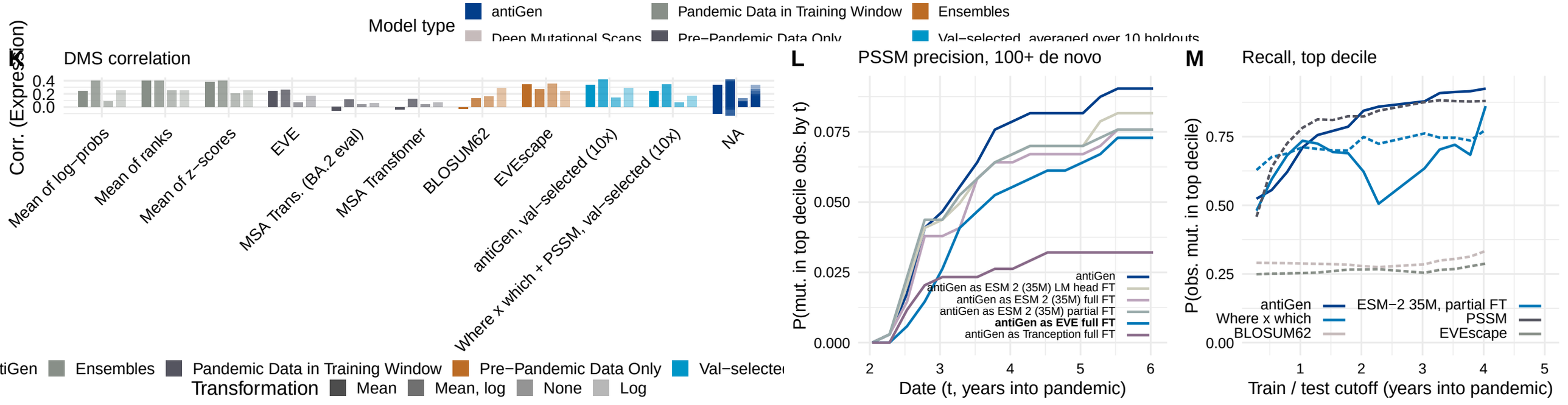
F
Definition of PSSM precision
Testing data downstream of query
F2Y
L5V
V3A
PSSM downstream of query
L8V F4W
Query (VOI)
M F V W L V L V
Training tree
antiGen
1 2 3 4 5 6 7 8
A
C
D
E
ℙ[prediction in top decile appears at frequency ≥ M]
F2Y
W4F
Top predicted decile
Time


G
PSSM precision, all mutations
P(mut. in top decile obs. by t)
Date (t, years into pandemic)
Model
antiGen
BA.2 DMS (LY−CoV1404 escape)
PSSM
Codon table
Mean of ranks
antiGen, val−selected (10x)


H
PSSM precision, de novo
P(mut. in top decile obs. by t)
Date (t, years into pandemic)
Model
antiGen
BA.2 DMS (expression)
Where × which
Codon table
Mean of ranks
antiGen, val−selected (10x)


I
Max precision, all mutations
Where x which + PSSM, val−selec
antiGen, val−selec
Mean of
Mean of lo
Mean
BA.2 DMS (LY−CoV1404
BA.2 DMS (ex
BA.2 DM
BA.1 DMS (ex
BA.1 DM
Where
Substitutio
Tra
MSA Trans. (B
MSA Tra
PSSM Cons
Co
BL
ESM
ESM
E
2nd
3rd
1st
0.00
0.25
0.50
0.75
Max PSSM precision


J
Max precision, de novo
1st
2nd
3rd
0.0
0.2
0.4
0.6
Max PSSM precision


Model type
antiGen
Pandemic Data in Training Window
Ensembles
Deep Mutational Scans
Pre−Pandemic Data Only
Val−selected, averaged over 10 holdouts
K
DMS correlation
Corr. (Expression)
Mean of log−probs
Mean of ranks
Mean of z−scores
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
BLOSUM62
EVEscape
antiGen, val−selected (10x)
Where x which + PSSM, val−selected (10x)
NA
Transformation
Mean
Mean, log
None
Log
L
PSSM precision, 100+ de novo
P(mut. in top decile obs. by t)
Date (t, years into pandemic)
antiGen
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (35M) partial FT
antiGen as EVE full FT
antiGen as Tranception full FT
M
Recall, top decile
P(obs. mut. in top decile)
Train / test cutoff (years into pandemic)
antiGen
Where x which
BLOSUM62
ESM−2 35M, partial FT
PSSM
EVEscape